

12.3 Revisiting Principal Component Analysis



The core skills you'll master in this section:

- ▶ Determine whether standardization is needed, based on whether the standard deviations across features differ excessively.
- ▶ Compute the correlation coefficient matrix after standardization.
- ▶ Eigenvalue decomposition of the correlation coefficient matrix.
- ▶ The first principal component explains the most variance.
- ▶ Distinguish between loadings and factor scores.
- ▶ The proportion of variance explained by each principal component, and the cumulative proportion, used to evaluate the effectiveness of dimensionality reduction.
- ▶ The operating steps and geometric meaning of using the first p principal components to approximately reconstruct the original data.

The previous section discussed principal component analysis from the perspective of an ellipse; we trust you now understand that, thinking of the data as a rotated ellipse, principal component analysis is about finding the right directions to straighten out that rotated ellipse. This section uses real data to revisit principal component analysis together, from an applied perspective.

Data

Figure 1 shows the day-over-day interest rate change data across different maturities (e.g., 0.5-year, 1-year, 2-year, ...).

⚠ Note that, Figure 1's data, multiplied by 100%, gives the day-over-day percentage change.



LA_12_03_01.ipynb is this section's companion code — every computation and visualization is carried out there. As you read this section, please follow along and practice in JupyterLab.

Looking at LA_12_03_01.ipynb, you'll find that the data matrix X has 248 rows and 8 columns — the data matrix X is a tall, thin matrix.

The data matrix X 's every row (a row vector) represents the change between the current day and the previous day; X 's every column represents one maturity.

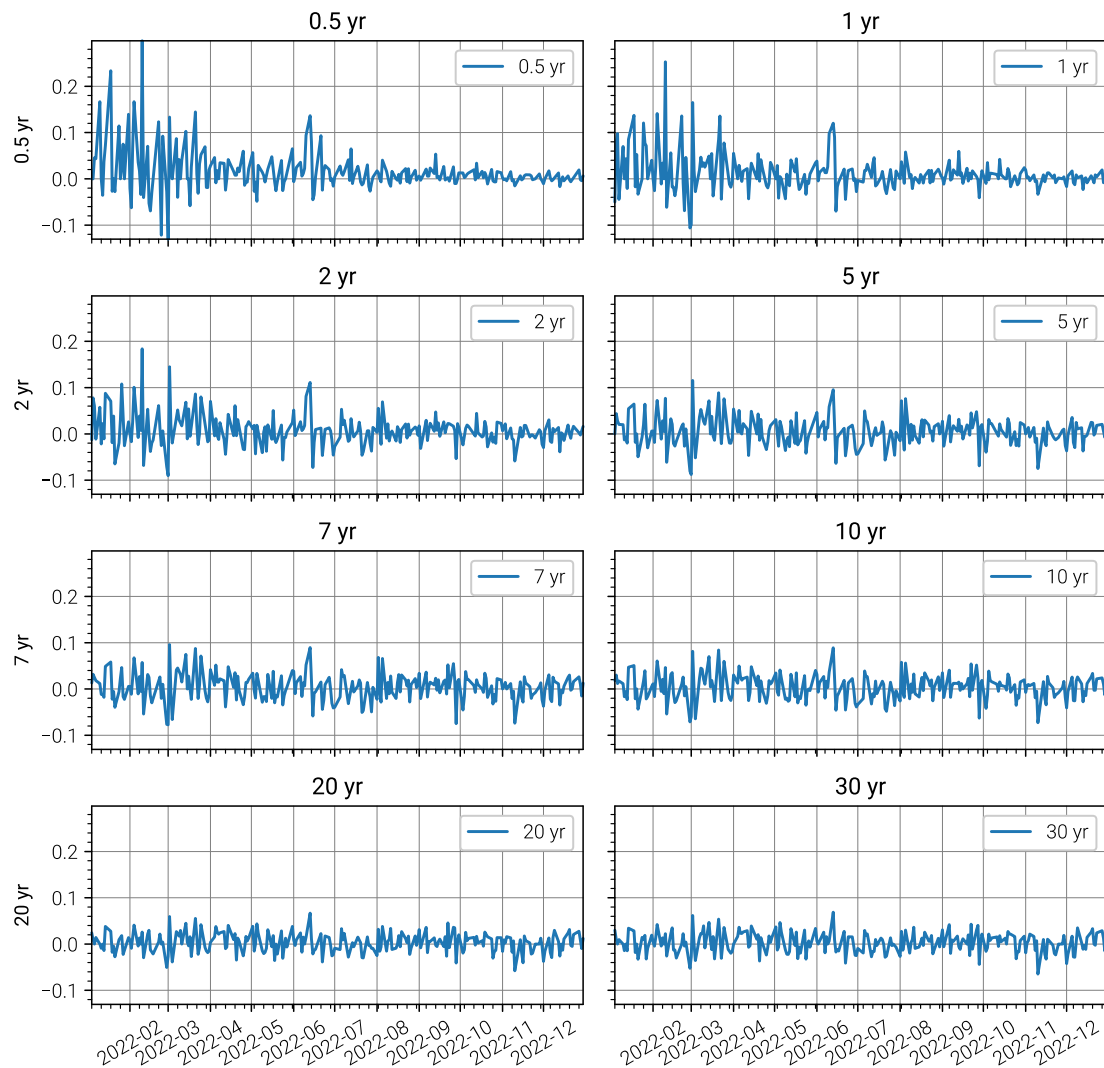


Figure 1. Line Plot of Day-over-Day Interest Rate Changes Across Different Maturities

Figure 1: it's hard to see whether there's some trend across the different maturities' changes, so we also plotted Figure 2. From Figure 2, this pairwise scatter plot, we can see that the interest rate changes across different maturities exhibit a strong positive linear correlation.

Next, let's try using principal component analysis to uncover the underlying pattern.

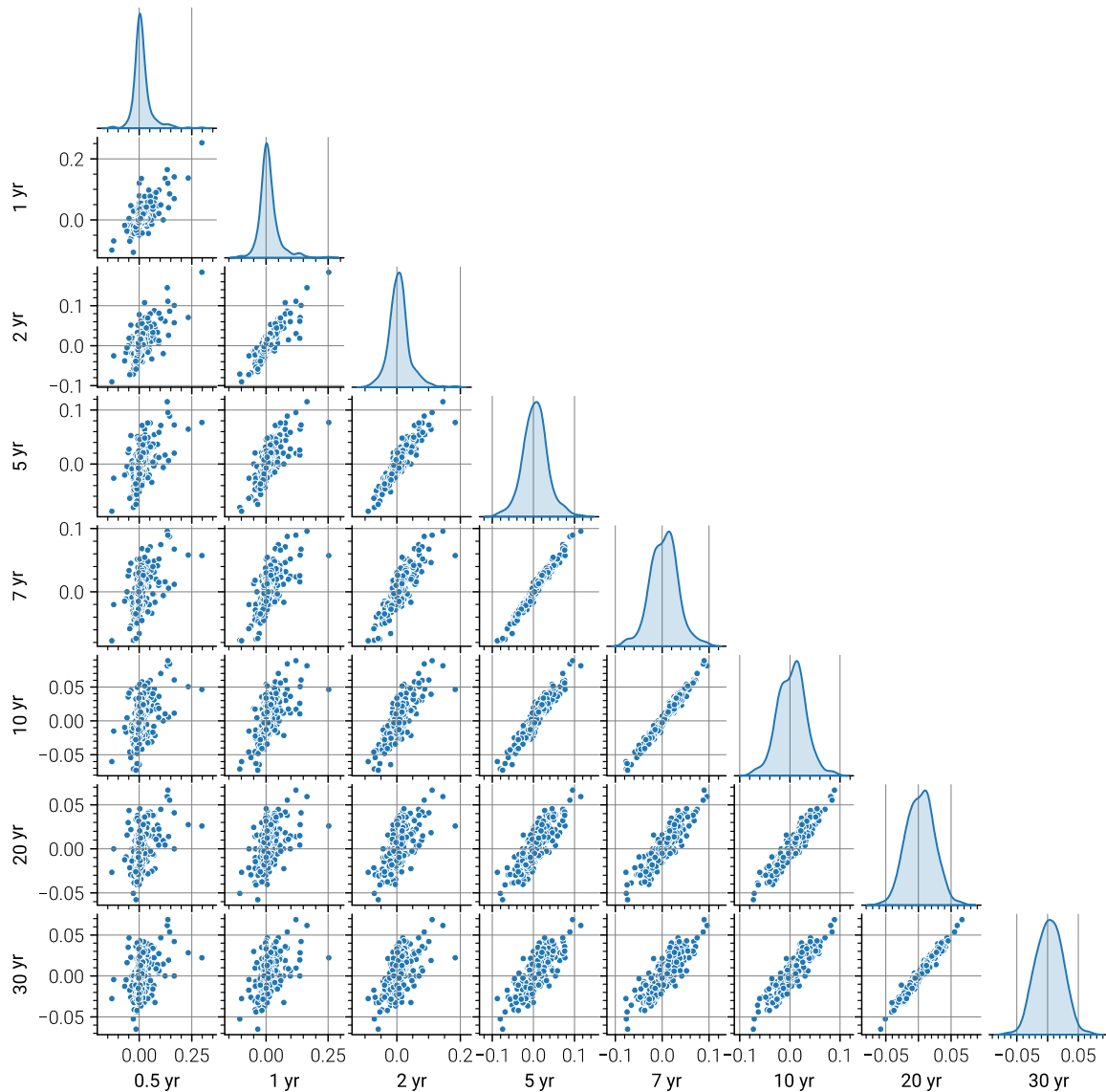


Figure 2. Pairwise Scatter Plot of Day-over-Day Interest Rate Changes Across Different Maturities, Omitting the Subplots Above the Main Diagonal

Covariance Matrix

The data matrix $\mathbf{X}$, after centering, becomes $\mathbf{X}_c$

$$\mathbf{X}_c = \mathbf{X} - \boldsymbol{\mu}^T$$

where $\boldsymbol{\mu}$ is $\mathbf{X}$'s centroid.

As mentioned earlier in this chapter, the covariance matrix is equivalent to $\mathbf{X}_c$'s special Gram matrix

$$\boldsymbol{\Sigma} = \frac{\mathbf{X}_c^T \mathbf{X}_c}{n-1}$$

Figure 3 shows the data matrix's covariance matrix.

Section 1 of this chapter explained that the covariance matrix's main-diagonal elements are variances, whose square roots are standard deviations; the covariance matrix's off-diagonal elements are covariances.

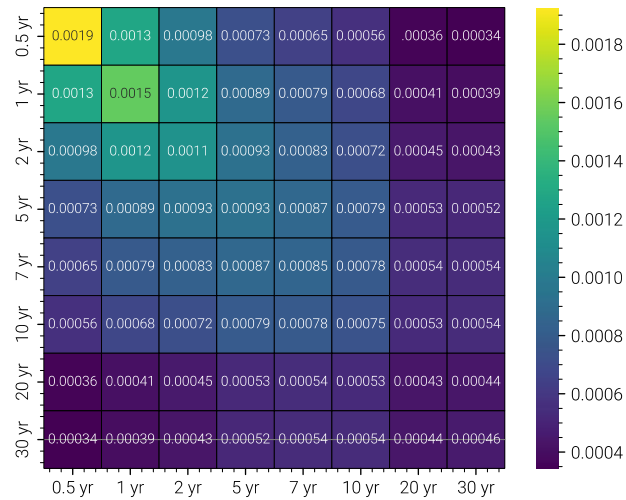


Figure 3. Heatmap of the Covariance Matrix of the Interest Rate Change Data

As shown in Figure 4, the standard deviations (volatility) of the interest rate change data differ greatly across maturities. As mentioned in the previous section, when performing principal component analysis, if the standard deviations across features differ too much, the data must first be standardized.

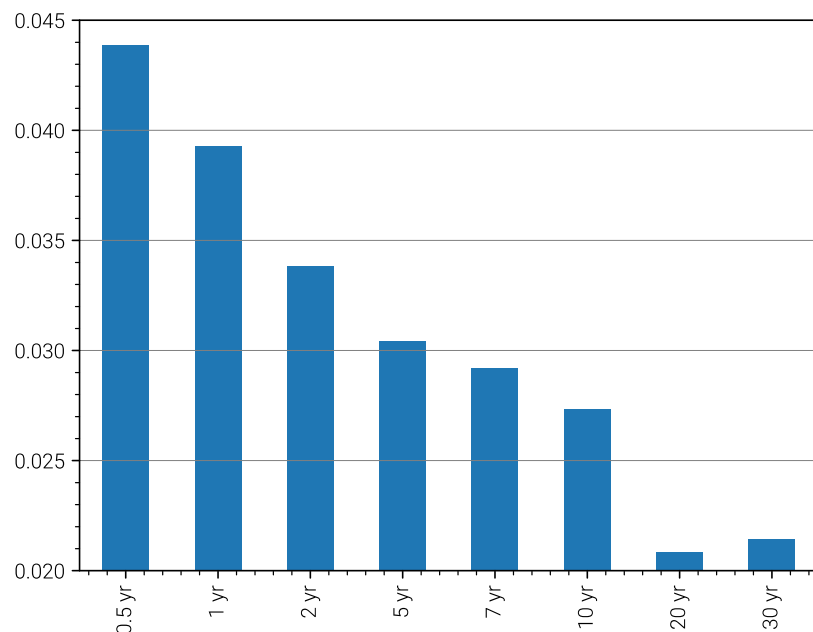


Figure 4. Bar Chart of Standard Deviations

Standardization

The standardized data matrix Z is

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$$\mathbf{Z} = \mathbf{X}_c \mathbf{D}^{-1} = (\mathbf{X} - \boldsymbol{\mu}^T) \mathbf{D}^{-1}$$

where $\mathbf{D}$ is a diagonal square matrix whose main-diagonal elements are the standard deviations

$$\mathbf{D} = \begin{bmatrix} \sigma_1 & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_D \end{bmatrix}$$

Figure 5 shows the line plot after standardization.

As mentioned earlier in this book, the key characteristic of standardized data is that it removes the original features' units, so that data with different dimensions and different scales can be compared on the same scale.

As shown in Figure 5, after standardization, each subplot's data has a mean of 0 and a standard deviation of 1.

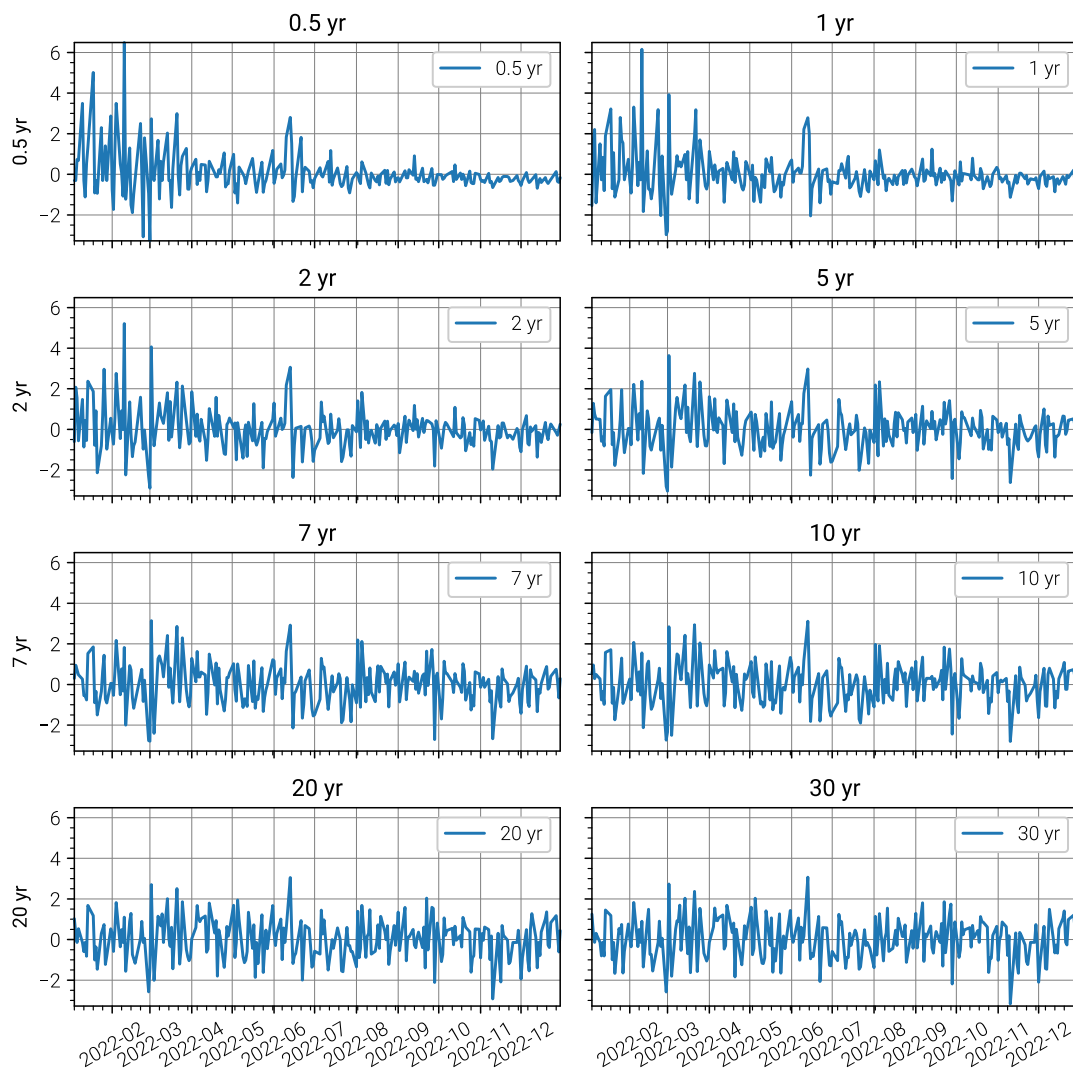


Figure 5. Line Plot of Standardized Data (Mean 0, Standard Deviation 1)

Figure 6 shows the pairwise scatter plot of the standardized data.

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Comparing Figure 2 and Figure 6, we find that standardization does not change the shape of the data's distribution, but it unifies the center location and dispersion, while preserving the relative positions and structure among the samples.

Standardized data no longer depends on the original units of measurement.

⚠ Note that standardization is fairly sensitive to outliers, because the mean and standard deviation themselves are easily influenced by extreme values.

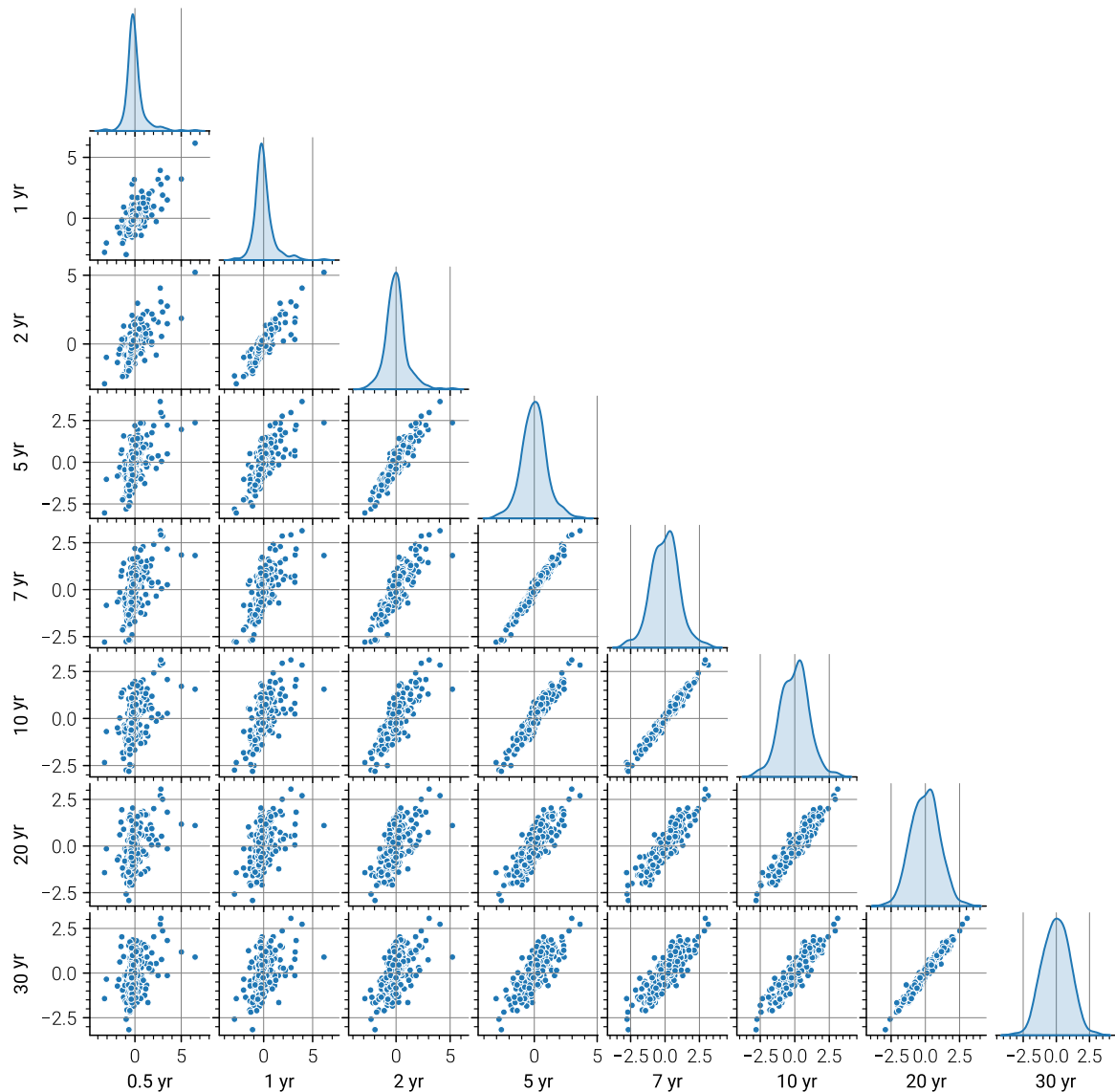


Figure 6. Pairwise Scatter Plot of Standardized Data, Omitting the Subplots Above the Main Diagonal

Linear Correlation Coefficient Matrix

As mentioned earlier, the linear correlation coefficient is a value used to measure the strength and direction of the linear relationship between two variables, ranging from -1 to 1 . A value of 1 indicates perfect positive correlation, -1 indicates perfect negative correlation, and 0 indicates no linear relationship.

The linear correlation coefficient reflects whether two variables' trends of change are consistent after removing units and standardizing, so it is unaffected by the variables' original units and scale.

As shown in Figure 7, the linear correlation coefficient matrix amounts to a summary of the pairwise linear correlation coefficients among multiple features

$$\mathbf{P} = \begin{bmatrix} 1 & \rho_{1,2} & \cdots & \rho_{1,D} \\ \rho_{1,2} & 1 & \cdots & \rho_{2,D} \\ \vdots & \vdots & \ddots & \vdots \\ \rho_{1,D} & \rho_{2,D} & \cdots & 1 \end{bmatrix}$$

Every element of the linear correlation coefficient matrix represents the degree of linear correlation between the corresponding pair of features. The values on the diagonal are always 1, because every feature is perfectly positively correlated with itself.

The linear correlation coefficient matrix is equivalent to the covariance matrix of the standardized data, that is

$$\mathbf{P} = \boldsymbol{\Sigma}_z = \frac{\mathbf{Z}^T \mathbf{Z}}{n-1}$$

Gram matrix

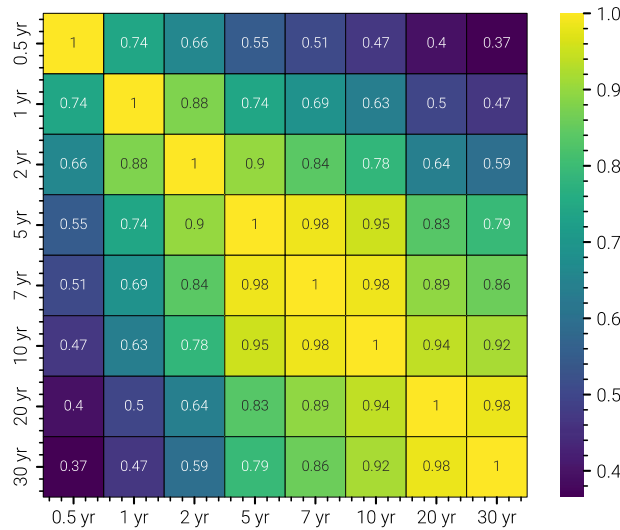


Figure 7. Linear Correlation Coefficient Matrix

From the perspective of orthogonal projection, projecting $\mathbf{Z}$ onto $\mathbf{e}_1$ gives the data $\mathbf{z}_1$, which is

$$\mathbf{z}_1 = \mathbf{Z}\mathbf{e}_1 = \begin{bmatrix} \mathbf{z}_1 & \mathbf{z}_2 & \cdots & \mathbf{z}_D \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

Computing $\mathbf{z}_1$'s variance

$$\text{var}(\mathbf{z}_1) = \frac{\mathbf{z}_1^T \mathbf{z}_1}{n-1} = \frac{(\mathbf{Z}\mathbf{e}_1)^T \mathbf{Z}\mathbf{e}_1}{n-1} = \mathbf{e}_1^T \frac{\mathbf{Z}^T \mathbf{Z}}{n-1} \mathbf{e}_1 = \mathbf{e}_1^T \mathbf{P} \mathbf{e}_1 = 1$$

The equation above amounts to extracting $\mathbf{P}$'s row 1, column 1 element, which means $\mathbf{z}_1$'s variance is 1.

⚠ Note that $\mathbf{Z}$'s every column is already centered (mean-removed), so when computing the variance, there's no need to repeat the centering.

Similarly, projecting $\mathbf{Z}$ onto $\mathbf{e}_2$ gives the data $\mathbf{z}_2$

$$\mathbf{z}_2 = \mathbf{Z}\mathbf{e}_2$$

$\mathbf{z}_1$ and $\mathbf{z}_2$'s covariance is

$$\text{cov}(\mathbf{z}_1, \mathbf{z}_2) = \text{cov}(\mathbf{z}_2, \mathbf{z}_1) = \frac{\mathbf{z}_2^T \mathbf{z}_1}{n-1} = \frac{(\mathbf{Z}\mathbf{e}_2)^T \mathbf{Z}\mathbf{e}_1}{n-1} = \mathbf{e}_2^T \frac{\mathbf{Z}^T \mathbf{Z}}{n-1} \mathbf{e}_1 = \mathbf{e}_2^T \mathbf{P} \mathbf{e}_1 = \rho_{1,2}$$

Eigenvalue Decomposition

Figure 8 shows the eigenvalue decomposition of the linear correlation coefficient matrix $\mathbf{P}$, that is

$$\mathbf{P} = \mathbf{V}\mathbf{A}\mathbf{V}^T$$

The linear correlation coefficient matrix is symmetric, so its eigenvalue decomposition is a spectral decomposition. In Figure 8, $\mathbf{V}$ is an orthogonal matrix, that is, an orthonormal basis.

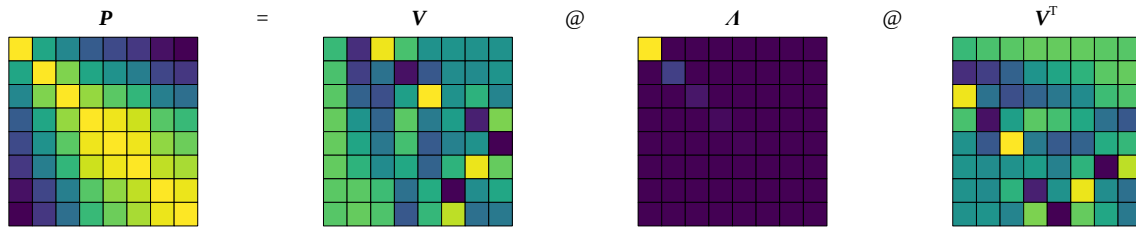


Figure 8. Eigenvalue Decomposition of the Linear Correlation Coefficient Matrix

The linear correlation coefficient matrix $\mathbf{P}$'s diagonalization can be written as

$$\mathbf{A} = \mathbf{V}^T \mathbf{P} \mathbf{V}$$

Writing $\mathbf{V}$ expanded in terms of its row vectors, gives

$$\begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_D \end{bmatrix} = \begin{bmatrix} \mathbf{v}_1^T \mathbf{P} \mathbf{v}_1 & \mathbf{v}_1^T \mathbf{P} \mathbf{v}_2 & \cdots & \mathbf{v}_1^T \mathbf{P} \mathbf{v}_D \\ \mathbf{v}_2^T \mathbf{P} \mathbf{v}_1 & \mathbf{v}_2^T \mathbf{P} \mathbf{v}_2 & \cdots & \mathbf{v}_2^T \mathbf{P} \mathbf{v}_D \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{v}_D^T \mathbf{P} \mathbf{v}_1 & \mathbf{v}_D^T \mathbf{P} \mathbf{v}_2 & \cdots & \mathbf{v}_D^T \mathbf{P} \mathbf{v}_D \end{bmatrix}$$

From the perspective of orthogonal projection, projecting $\mathbf{Z}$ onto $\mathbf{v}_1$ gives the data $\mathbf{y}_1$, which is

$$\mathbf{y}_1 = \mathbf{Z}\mathbf{v}_1$$

and $\mathbf{y}_1$'s variance is

$$\text{var}(\mathbf{y}_1) = \frac{\mathbf{y}_1^T \mathbf{y}_1}{n-1} = \frac{(\mathbf{Z}\mathbf{v}_1)^T \mathbf{Z}\mathbf{v}_1}{n-1} = \mathbf{v}_1^T \frac{\mathbf{Z}^T \mathbf{Z}}{n-1} \mathbf{v}_1 = \mathbf{v}_1^T \mathbf{P} \mathbf{v}_1 = \lambda_1$$

In other words, in the direction of $\mathbf{v}_1$ (the first principal component direction), the variance of $\mathbf{Z}$'s projection is maximized; that is, $\mathbf{v}_1$ explains the most variance in the data.

In principal component analysis, $\mathbf{V}$'s column vectors are also called loadings (loading).

Note that in much of the literature, an alternative definition of loading multiplies $\mathbf{V}$'s column vectors by the square root of the corresponding principal component's eigenvalue; for example, the first principal component's loading is

$$\mathbf{v}_1 \sqrt{\lambda_1}$$

Factor score (score) represents each sample's projected coordinates in the principal-component coordinate system, that is, $\mathbf{y}$.

The matrix formed by the factor scores corresponds to the following matrix multiplication

$$\mathbf{Y} = \mathbf{ZV}$$

That is, orthogonally projecting $\mathbf{Z}$ onto $\mathbf{V}$ gives $\mathbf{Y}$.

 Please draw the matrix-operation heatmap corresponding to the equation above.

Writing $\mathbf{V}$ as column vectors and expanding, gives

$$[\mathbf{y}_1 \ \mathbf{y}_2 \ \cdots \ \mathbf{y}_D] = [\mathbf{Zv}_1 \ \mathbf{Zv}_2 \ \cdots \ \mathbf{Zv}_D]$$

and $\mathbf{Y}$'s covariance matrix is exactly the diagonal square matrix obtained from the spectral decomposition, that is

$$\Sigma_Y = \frac{\mathbf{Y}^T \mathbf{Y}}{n-1} = \frac{(\mathbf{ZV})^T \mathbf{ZV}}{n-1} = \mathbf{V}^T \frac{\mathbf{Z}^T \mathbf{Z}}{n-1} \mathbf{V} = \mathbf{V}^T \mathbf{P} \mathbf{V} = \mathbf{I}$$

 Please draw the matrix-operation heatmap corresponding to the equation above.

Eigenvalues

The linear correlation coefficient matrix $\mathbf{P}$'s main-diagonal sum is $\mathbf{P}$'s trace, that is, $\text{trace}(\mathbf{P}) = 8$. This is equivalent to the total variance of the standardized data $\mathbf{Z}$.

As mentioned earlier, $\mathbf{A}$'s trace is also 8, that is

$$\text{trace}(\mathbf{A}) = \lambda_1 + \lambda_2 + \cdots + \lambda_8 = 8$$

Figure 9 shows the eigenvalues arranged from largest to smallest.

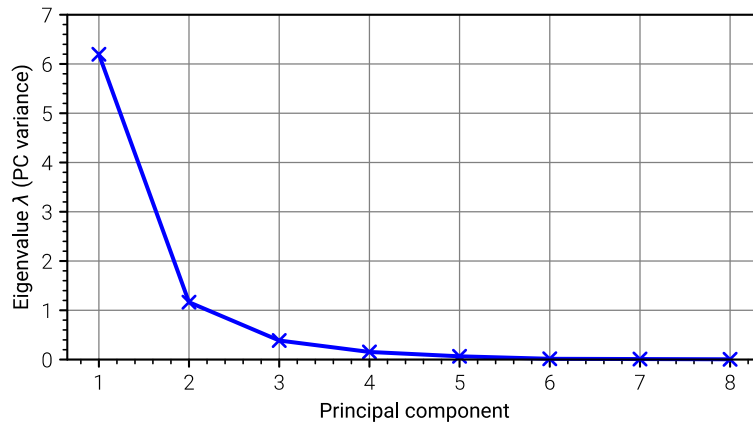


Figure 9. Eigenvalues Arranged from Largest to Smallest

The j -th eigenvalue λ_j 's contribution percentage to the total variance is:

$$\frac{\lambda_j}{\sum_{j=1}^D \lambda_j} 100\%$$

The equation above's denominator is the total variance of the data, that is, $\text{trace}(\mathbf{P}) = \text{trace}(\mathbf{A})$.

This ratio can be used to measure the j -th principal component's ability to explain the data.

The cumulative percentage of explained variance from the first p eigenvalues (cumulative percentage of explained variance) is:

$$\frac{\sum_{j=1}^p \lambda_j}{\sum_{j=1}^D \lambda_j} 100\%$$

As shown in Figure 10, this percentage represents the proportion that the sum of variance explained by the first p principal components makes up of the sum of variance explained by all principal components. The cumulative percentage of explained variance can be used to evaluate PCA's dimensionality-reduction effectiveness — it measures the proportion of the data's variance that the first p principal components can explain.

Typically, we hope to choose an appropriate number of principal components p so that the cumulative percentage of explained variance reaches a preset threshold (e.g., 80% or 90%), in order to retain as much of the original data's information as possible.

We find that the first principal component already explains nearly 80% of the total variance,

$$\frac{\lambda_1}{\lambda_1 + \lambda_2 + \dots + \lambda_8} \times 100\% = 77.477\%$$

while the first two principal components together explain over 90% of the total variance

$$\frac{\lambda_1 + \lambda_2}{\lambda_1 + \lambda_2 + \dots + \lambda_8} \times 100\% = 92.038\%$$

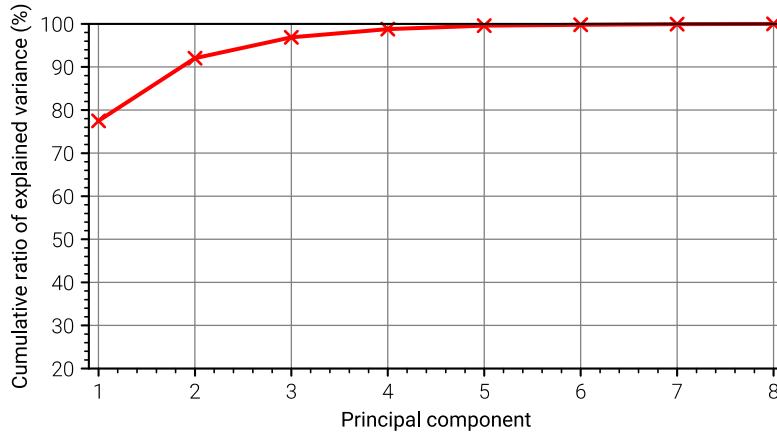


Figure 10. Cumulative Percentage of Total Variance Explained by the First p Eigenvalues

Approximate Reconstruction

If we use only the first principal component's information to approximately reconstruct $\mathbf{Z}$ (not yet $\mathbf{X}$), this corresponds to the matrix multiplication

$$\hat{\mathbf{Z}} = \mathbf{Z}_1 = \mathbf{y}_1 @ \mathbf{v}_1^T = \mathbf{Z} @ (\mathbf{v}_1 @ \mathbf{v}_1^T)$$

The equation above amounts to the orthogonal projection of $\mathbf{Z}$ purely along the $\mathbf{v}_1$ direction.

$\mathbf{Z}$ and $\mathbf{Z}_1$'s error is

$$\mathbf{Z} - \mathbf{Z}_1 = \mathbf{Z} - \mathbf{Z} @ (\mathbf{v}_1 @ \mathbf{v}_1^T) = \mathbf{Z} (\mathbf{I} - \mathbf{v}_1 @ \mathbf{v}_1^T)$$

This is also an orthogonal projection.

In order to use $\mathbf{Z}_1$ to approximately reconstruct $\mathbf{X}$, we also need "scaling $\rightarrow$ translation"

$$\hat{\mathbf{X}} = \mathbf{Z}_1 \mathbf{D} + \boldsymbol{\mu}^T = (\mathbf{y}_1 @ \mathbf{v}_1^T) \mathbf{D} + \boldsymbol{\mu}^T = (\mathbf{Z} @ (\mathbf{v}_1 @ \mathbf{v}_1^T)) \mathbf{D} + \boldsymbol{\mu}^T$$

That is, the equation above represents:

- ▶ $\mathbf{Z}_1$ approximately reconstructs $\mathbf{Z}$ (the standardized data);
- ▶ Then scale via the diagonal square matrix $\mathbf{D}$ (whose diagonal elements are the standard deviations) to approximately reconstruct the centered data $\mathbf{X}_c$;
- ▶ Finally, translate using the centroid $\boldsymbol{\mu}^T$ (the broadcasting rule) to approximate the original data matrix $\mathbf{X}$.

Substituting the preceding result into the equation above gives

$$\hat{\mathbf{X}} = ((\mathbf{X} - \boldsymbol{\mu}^T) \mathbf{D}^{-1} @ (\mathbf{v}_1 @ \mathbf{v}_1^T)) \mathbf{D} + \boldsymbol{\mu}^T$$

The geometric operations corresponding to the equation above are "translation $\rightarrow$ scaling $\rightarrow$ projection $\rightarrow$ scaling $\rightarrow$ translation". Specifically,

- ▶ Translation: centering the data.
- ▶ Scaling: standardizing the data.
- ▶ Projection: projecting the standardized data onto the first principal component $\mathbf{v}_1$.
- ▶ Scaling: the inverse of the second step.

► Translation: the inverse of the first step.

Using the first principal component's information to approximately reconstruct $\mathbf{X}$'s error is

$$\mathbf{X} - \hat{\mathbf{X}} = \mathbf{X} - \left((\mathbf{X} - \boldsymbol{\mu}^T) \mathbf{D}^{-1} @ (\mathbf{v}_1 @ \mathbf{v}_1^T) \right) \mathbf{D} + \boldsymbol{\mu}^T$$

Figure 11: the blue curve is the original data, the orange curve is the approximate original data reconstructed from the first principal component's information, and black represents the error.

Figure 12 shows the pairwise scatter plot of the data approximately reconstructing $\mathbf{X}$ using the first principal component's information; the points in every subplot lie on a single line, because we're reconstructing the original data using just one principal component's information.

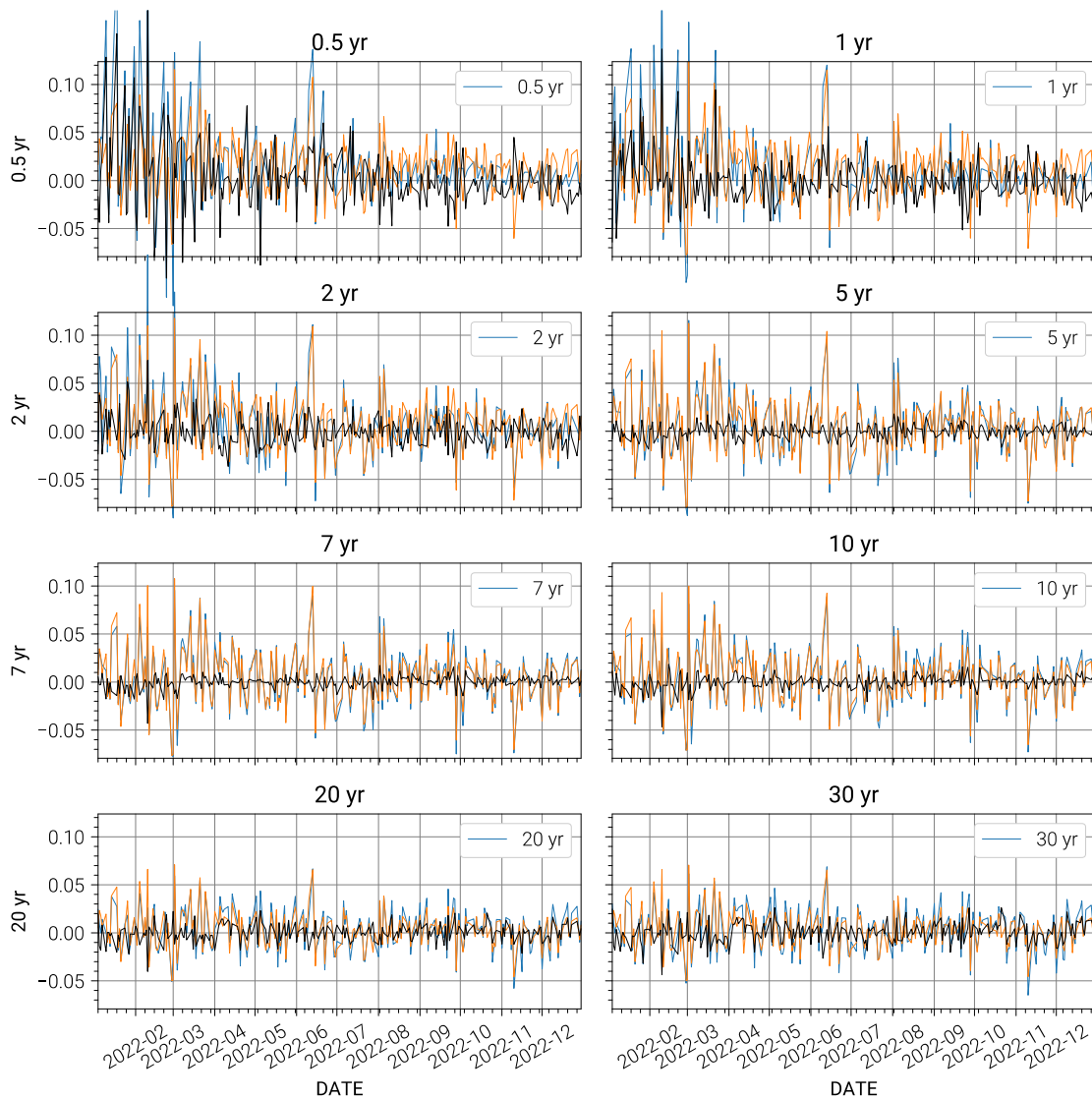


Figure 11. Comparing the Original Data and the Approximately Reconstructed Data via a Line Plot, First Principal Component

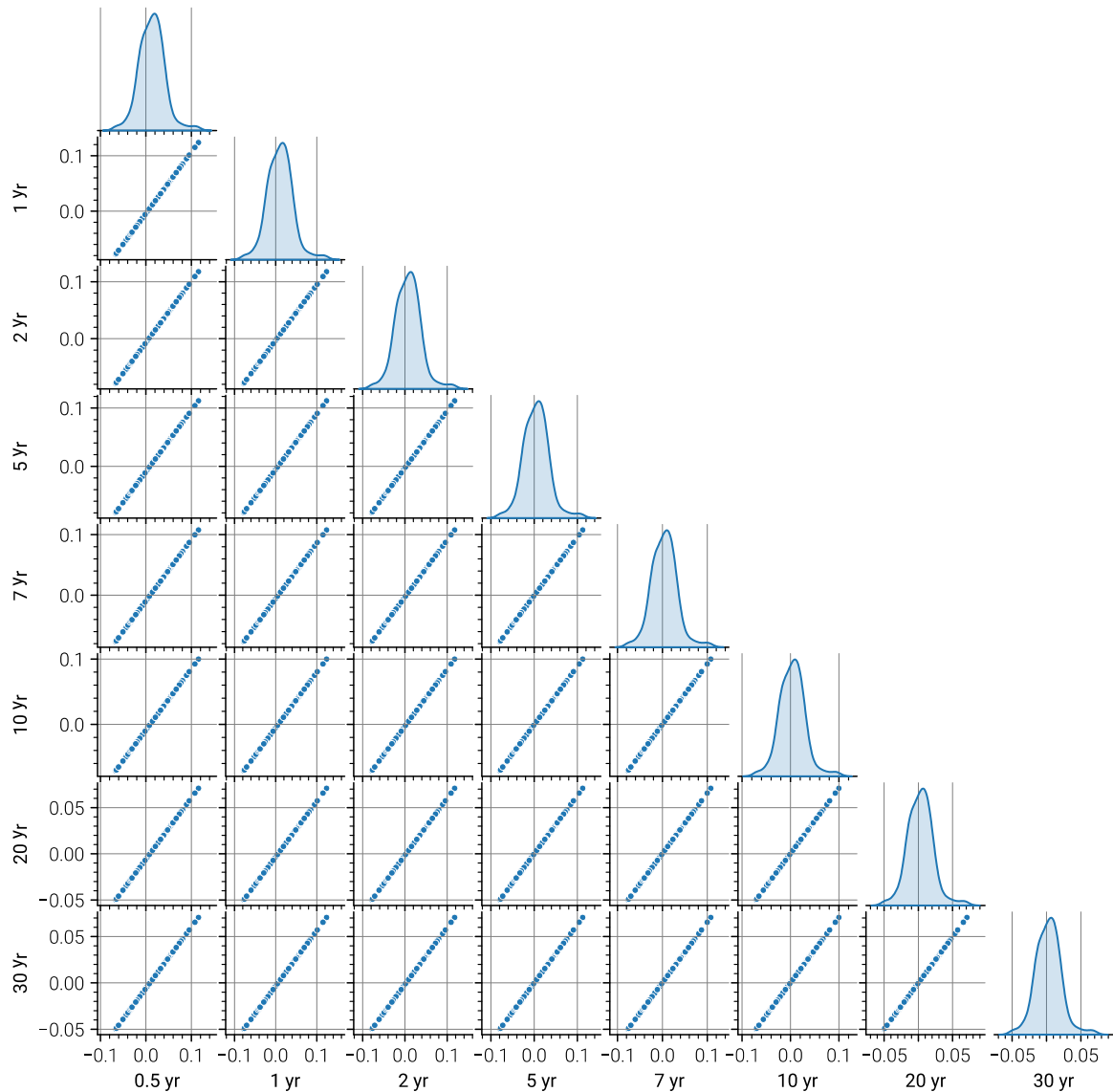


Figure 12. Pairwise Scatter Plot of the Data Approximately Reconstructed from the First Principal Component, Omitting the Subplots Above the Main Diagonal

To compare how similar the original data and the approximately reconstructed data are, we plot the scatter plot shown in Figure 13.

Figure 13 in this figure, the approximately reconstructed data is on the horizontal axis and the original data is on the vertical axis. The better the reconstruction, the closer each point's horizontal and vertical coordinates are — in other words, the closer they lie to a diagonal line $x_2 = x_1$. So, we add the reference line $x_2 = x_1$ (marked in red) to the plot, as the standard line for "perfect reconstruction". If most of the scatter points cluster near this red line, it indicates that the reconstruction closely resembles the original data. This graphical approach lets us judge the reconstruction quality intuitively.

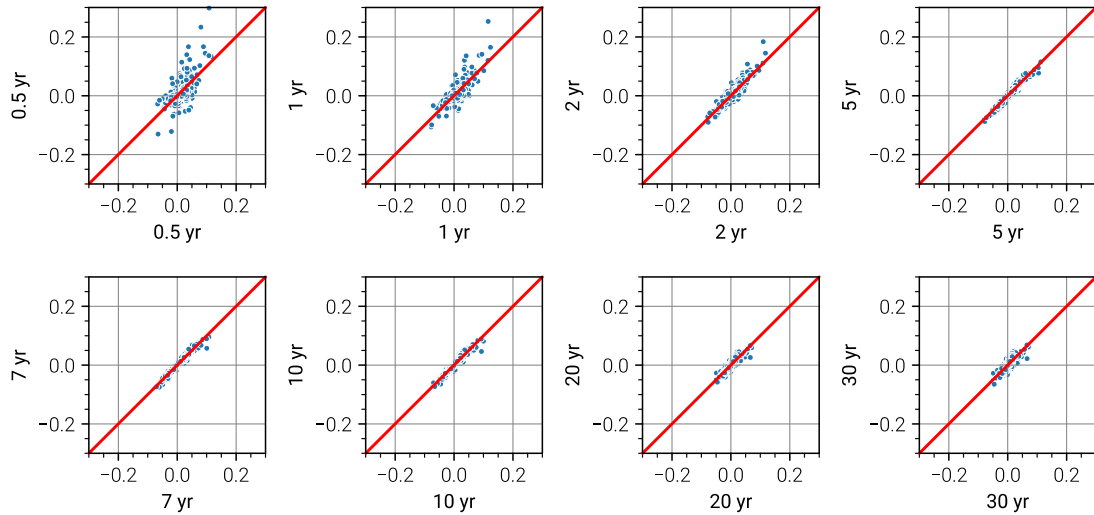


Figure 13. Comparing the Original Data and the Approximately Reconstructed Data via a Scatter Plot, First Principal Component

Using the first two principal components' information to approximately reconstruct $\mathbf{X}$, the corresponding matrix operation is

$$\hat{\mathbf{X}} = (\mathbf{Z}_1 + \mathbf{Z}_2) \mathbf{D} + \boldsymbol{\mu}^T = (\mathbf{y}_1 @ \mathbf{v}_1^T + \mathbf{y}_2 @ \mathbf{v}_2^T) \mathbf{D} + \boldsymbol{\mu}^T = \mathbf{Z} @ (\mathbf{v}_1 @ \mathbf{v}_1^T + \mathbf{v}_2 @ \mathbf{v}_2^T) \mathbf{D} + \boldsymbol{\mu}^T$$

? Please think about the geometric operation corresponding to the equation above. Based on it, compute $\mathbf{X}$'s reconstruction error using the first two principal components' information.

Figure 14, Figure 15, and Figure 16 show visualizations of approximately reconstructing the original data using the "first + second" principal components; we'll leave the analysis to you.

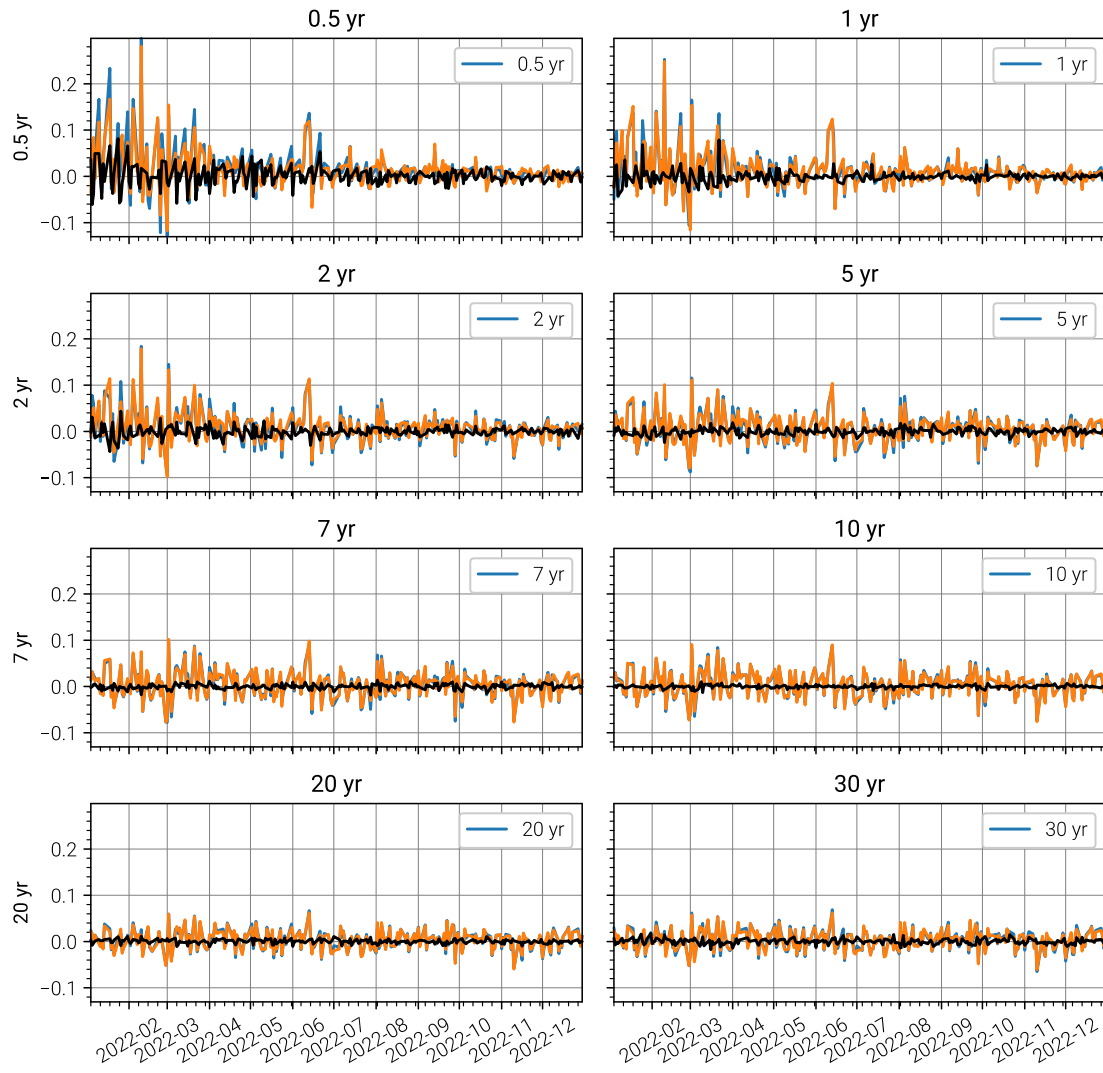


Figure 14. Comparing the Original Data and the Approximately Reconstructed Data via a Line Plot, "First + Second" Principal Components

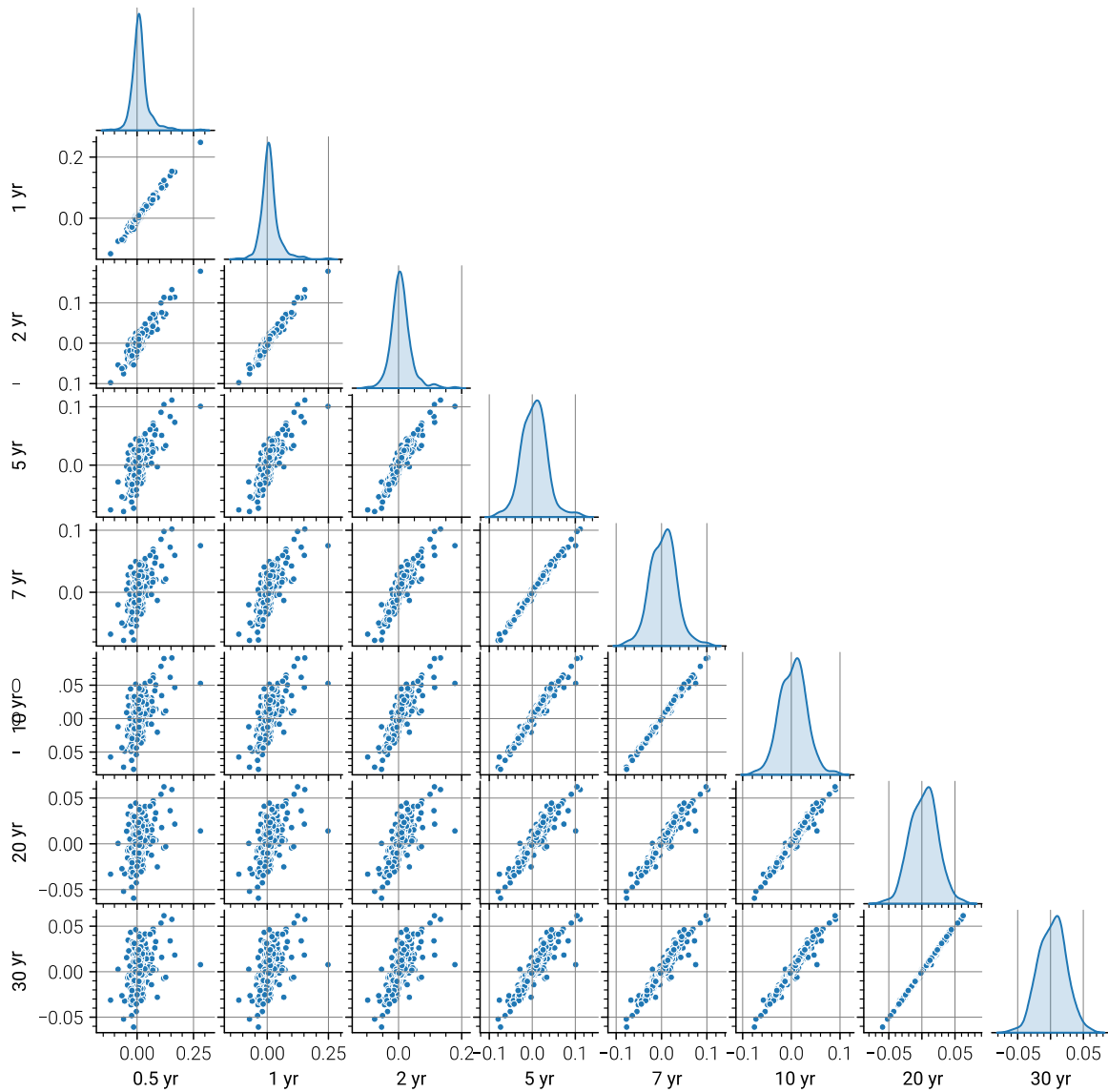
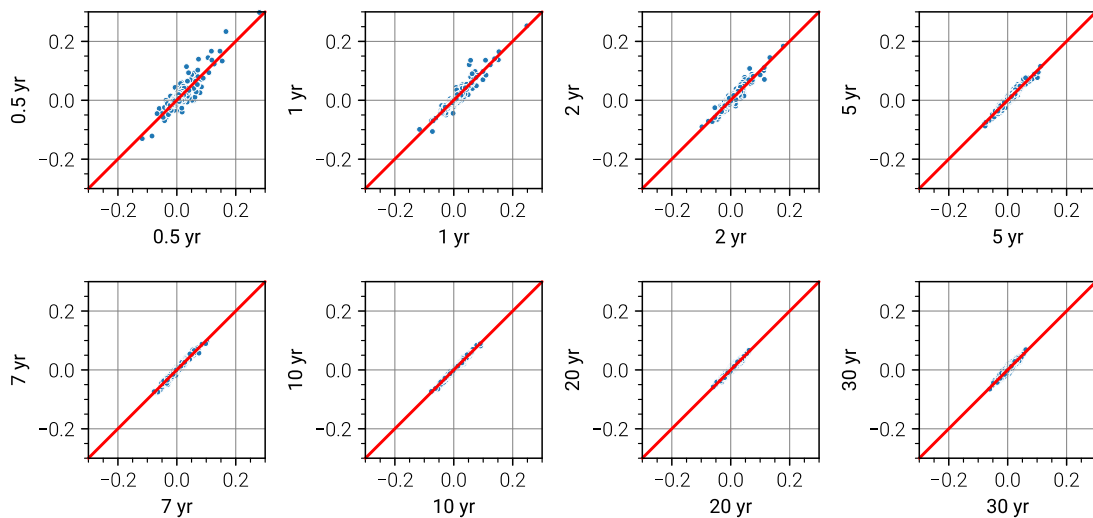


Figure 15. Pairwise Scatter Plot of the Data Approximately Reconstructed from the First and Second Principal Components, Omitting the Subplots Above the Main Diagonal



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Figure 16. Comparing the Original Data and the Approximately Reconstructed Data via a Scatter Plot, "First + Second" Principal Components



LA_12_03_02.ipynb uses `statsmodels.multivariate.pca()` to perform principal component analysis.

LA_12_03_01.ipynb essentially replicates LA_12_03_02.ipynb's results. We encourage you to study LA_12_03_02.ipynb on your own. Please note the sign differences in the eigenvectors.



Please use tools such as DeepSeek/ChatGPT to complete the following exercises for this section.

Q1. Please modify LA_12_03_01.ipynb to approximately reconstruct the original data using the first 3 and 4 principal components' information.

Q2. Please apply all of LA_12_03_01.ipynb's visualization schemes to LA_12_03_02.ipynb.

Q3. Please learn to use Scikit-learn's PCA function to complete this section's example:

<https://scikit-learn.org/stable/modules/generated/sklearn.decomposition.PCA.html>